

Algebraic tunneling for the restricted fractional Laplacian on distant domains

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Abstract. Let $D \subset B_R(0) \subset \mathbb{R}^d$ be a bounded centrally symmetric Lipschitz domain, and let Ω_L be the union of two centrally inverted translates of D whose centers are separated by L . For the restricted fractional Laplacian $(-\Delta)^s$, $0 < s < 1$, let μ_1 be the one-well ground-state eigenvalue, let ϕ_1 be its positive L^2 -normalized eigenfunction, and put $m_1 = \int_D \phi_1$. As $L \rightarrow \infty$,

$$\begin{aligned}\lambda_1(\Omega_L) &= \mu_1 - c_{d,s} m_1^2 L^{-d-2s} + O(L^{-d-2s-2} + L^{-2d-4s}), \\ \lambda_2(\Omega_L) &= \mu_1 + c_{d,s} m_1^2 L^{-d-2s} + O(L^{-d-2s-2} + L^{-2d-4s}),\end{aligned}$$

where $c_{d,s}$ is the singular-integral normalization. For two equal balls, the second formula gives the leading rate in the nonlocal Hong–Krahn–Szegő minimizing sequence. More generally, for any bounded Lipschitz reference domain and fixed distinct translation sites $a_1, \dots, a_N$, the first N eigenvalues are

$$\lambda_k = \mu_1 + L^{-d-2s} \theta_k + O(L^{-d-2s-2} + L^{-2d-4s}), \quad 1 \leq k \leq N,$$

where $\theta_1 \leq \dots \leq \theta_N$ are the eigenvalues of the explicit matrix with zero diagonal and off-diagonal entries $-c_{d,s} m_1^2 |a_i - a_j|^{-d-2s}$. Exact cell-integral Galerkin computations on two and three intervals illustrate these fixed-mesh limits.

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1. Introduction

The restricted fractional Laplacian couples disjoint components through its integral kernel, unlike the local Dirichlet Laplacian, whose spectrum on a disconnected union is exactly the multiset union of the component spectra. We prove that two distant congruent components exhibit a nonzero splitting that decays algebraically with their separation.

Brasco and Parini proved that the fractional Hong–Krahn–Szegő bound for the second eigenvalue is strict and that two equal balls form a minimizing sequence when their mutual distance tends to infinity [BP16, Theorem 6.2]. Mixed-operator analogues were

proved by Biagi, Dipierro, Valdinoci, and Vecchi [BDVV23, Theorem 1.1], and by Goel and Sreenadh for a local p -Laplacian coupled to a compact-kernel nonlocal p -Laplacian [GS19, Theorem 1.4]; the latter interaction is finite-range rather than algebraic. Parini and Salort placed such distant components in a general concentration–compactness dichotomy [PS20, Theorems 1.1 and 1.2]. For two one-dimensional patches, Léculier and Roquejoffre proved monotonicity and continuity of the principal eigenvalue throughout the separation range, including contact [LR23, Theorem 1], but did not identify its leading large-separation coefficient or the accompanying second-eigenvalue split. Together these works give qualitative limiting, continuity, and dichotomy results, but not the leading interaction coefficient for the standard restricted fractional Laplacian. We extract that coefficient and the associated effective finite-dimensional interaction.

The influence of mutual position on the restricted fractional spectrum is also emphasized by Abatangelo, Felli, and Noris [AFN20, Introduction, p. 3], in a study of small fractional-capacity perturbations. In the shape-optimization setting, Zahl isolated the exact cross-energy under translations and proposed a signed point-mass interaction with kernel $|x - y|^{-d-2s}$ [Zah26, Section 10, equations (10.3) and (10.7), and Conjecture 10.9]. Wu subsequently proved that this discrete signed-mass model has no local minima, settling that conjecture [Wu26, Introduction]. In another recent direction, Dipierro, Proietti Lippi, Sportelli, and Valdinoci studied disconnected domains for generalized eigenproblems having a superposition of lower-order operators on the right-hand side [DPLSV26, Theorems 1.4–1.7]. None of these works derives a large-separation spectral expansion for the standard restricted fractional Laplacian. Zahl’s discrete energy anticipates the matrix in (4.9); here it is derived from the full operator, with operator-norm control of the off-diagonal block and spectral-gap control of the complementary modes, and every ordered cluster eigenvalue is identified. We are not aware of an earlier result giving either the two-component coefficient with a quantified remainder or this finite-well spectral reduction.

The first result, Theorem 3.3, treats the double ground-state cluster. Theorem 4.6 identifies the entire ground-state cluster for finitely many translated components with the spectrum of an explicit weighted interaction matrix; unlike the central-inversion reduction, this result does not require central symmetry of the reference component. Its $N = 2$ specialization, Corollary 4.7, gives the same leading doublet coefficient without symmetry; symmetry is used only for the exact scalar-branch decomposition and the matching explicit remainder in Theorem 3.3. Both proofs are self-contained after the standard one-well spectral facts. The interval provides an explicit benchmark: one-well eigenvalue and eigenfunction estimates are available in [Kwa12, Theorem 1 and Propositions 1–2]; sharper eigenvalue asymptotics and an eigenfunction bound uniform in both the index and the fractional order are proved in [Zha26, Theorems 1 and 2]. Both sources write the operator order as $\alpha = 2s$.

2. The one-well operator and exact two-well reduction

Fix $d \geq 1$ and $0 < s < 1$, and put

$$\kappa = d + 2s.$$

For an open set $G \subset \mathbb{R}^d$, let

$$\tilde{H}^s(G) = \{u \in H^s(\mathbb{R}^d) : u = 0 \text{ a.e. on } \mathbb{R}^d \setminus G\}.$$

We use the Gagliardo seminorm

$$[u]_{H^s(\mathbb{R}^d)}^2 = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|u(x) - u(y)|^2}{|x - y|^{d+2s}} dx dy.$$

For bounded Lipschitz G , this space agrees with the completion of $C_c^\infty(G)$ in the global $H^s(\mathbb{R}^d)$ norm [DFW21, Section 2, equation (2.1)]. The zero-exterior quadratic form is

$$\Pi_G[u] = \frac{c_{d,s}}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|u(x) - u(y)|^2}{|x - y|^\kappa} dx dy, \quad u \in \tilde{H}^s(G), \quad (2.1)$$

where $c_{d,s} = 4^s s \Gamma(d/2 + s) / (\pi^{d/2} \Gamma(1 - s))$ is the standard normalization in the singular-integral definition of $(-\Delta)^s$. The form norm $(\Pi_G[u] + \|u\|_{L^2(G)}^2)^{1/2}$ is equivalent to the norm inherited from $H^s(\mathbb{R}^d)$ on $\tilde{H}^s(G)$. Thus (2.1) is a densely defined closed form, and we denote its associated self-adjoint operator on $L^2(G)$ by A_G . We write $\lambda_1(G) \leq \lambda_2(G) \leq \dots$ for the eigenvalues of A_G , repeated according to multiplicity. For an abstract self-adjoint operator T with compact resolvent, $\lambda_k(T)$ has the analogous meaning.

Let $D \subset B_R(0)$ be a bounded Lipschitz domain satisfying $D = -D$. We retain Lipschitz regularity in order to invoke the extension property and compactness of the form embedding from [DNPV12, Theorems 5.4 and 7.1]. Hence A_D has compact resolvent. Write its eigenvalues with multiplicity and denote the first two distinct levels by

$$\mu_1 < \mu_2, \quad g = \mu_2 - \mu_1 > 0.$$

The first eigenvalue is simple and has a strictly positive normalized eigenfunction by [BP16, Theorem 2.8], specialized to the linear case. Fix that eigenfunction and set

$$A_D \phi_1 = \mu_1 \phi_1, \quad \|\phi_1\|_{L^2(D)} = 1, \quad m_1 = \int_D \phi_1(x) dx > 0. \quad (2.2)$$

Simplicity and central-inversion invariance show that $\phi_1(-x) = \pm \phi_1(x)$; positivity fixes the sign, so $\phi_1(-x) = \phi_1(x)$.

Fix a unit vector $e \in \mathbb{R}^d$. For $L > 2R$, define

$$D_L^- = D - \frac{L}{2}e, \quad D_L^+ = D + \frac{L}{2}e, \quad \Omega_L = D_L^- \cup D_L^+.$$

On $L^2(D)$ define the bounded integral operator

$$(B_L v)(x) = -c_{d,s} \int_D \frac{v(y)}{|Le - x - y|^\kappa} dy. \quad (2.3)$$

Its kernel is real and symmetric, so B_L is self-adjoint.

Lemma 2.1 (*Exact central-inversion reduction*). *The operator A_{Ω_L} is unitarily equivalent to*

$$\begin{pmatrix} A_D & B_L \\ B_L & A_D \end{pmatrix} \quad \text{on } L^2(D) \oplus L^2(D),$$

and hence to

$$(A_D + B_L) \oplus (A_D - B_L). \quad (2.4)$$

Proof. For $f \in \tilde{H}^s(\Omega_L)$ set

$$u(x) = f\left(x - \frac{L}{2}e\right), \quad v(x) = f\left(-x + \frac{L}{2}e\right), \quad x \in D.$$

This defines a unitary map from $L^2(\Omega_L)$ to $L^2(D) \oplus L^2(D)$. Translation and central-inversion invariance identify the two diagonal parts of the form with $\Pi_D[u]$ and $\Pi_D[v]$.

To identify the form domains, fix $\chi \in C_c^\infty(\mathbb{R}^d)$ and $w \in H^s(\mathbb{R}^d)$. Then

$$\begin{aligned} [\chi w]_{H^s(\mathbb{R}^d)}^2 &\leq 2\|\chi\|_{L^\infty}^2 [w]_{H^s(\mathbb{R}^d)}^2 \\ &\quad + 2 \int_{\mathbb{R}^d} |w(y)|^2 \int_{\mathbb{R}^d} \frac{|\chi(x) - \chi(y)|^2}{|x - y|^{d+2s}} dx dy. \end{aligned}$$

The inner integral is bounded uniformly in y : use $|\chi(x) - \chi(y)| \leq \|\nabla \chi\|_{L^\infty} |x - y|$ when $|x - y| \leq 1$ and $|\chi(x) - \chi(y)| \leq 2\|\chi\|_{L^\infty}$ otherwise. The two radial bounds are integrable because $s < 1$ and $s > 0$, respectively. Consequently multiplication by χ is bounded on $H^s(\mathbb{R}^d)$.

Positive separation permits cutoffs $\chi_\pm$ that equal one on $D_L^\pm$ and vanish near the other component. For $f \in \tilde{H}^s(\Omega_L)$, the functions $\chi_- f$ and $\chi_+ f$ therefore belong to $H^s(\mathbb{R}^d)$ and, after the above changes of variables, give $u, v \in \tilde{H}^s(D)$. Conversely, translated and reflected zero extensions of any $u, v \in \tilde{H}^s(D)$ belong to $H^s(\mathbb{R}^d)$, and their sum is supported in Ω_L . Hence the unitary identifies $\tilde{H}^s(\Omega_L)$ with $\tilde{H}^s(D) \oplus \tilde{H}^s(D)$.

The cross-kernel is bounded because of positive separation. Each separate zero-exterior form already contains the square interaction with the location occupied by the other component. Thus, after the above changes of variables, expanding the square in (2.1) gives exactly

$$\begin{aligned} \Pi_{\Omega_L}[f] - \Pi_D[u] - \Pi_D[v] &= c_{d,s} \iint_{D \times D} \frac{|u(x) - v(y)|^2 - |u(x)|^2 - |v(y)|^2}{|Le - x - y|^\kappa} dx dy \\ &= -2c_{d,s} \operatorname{Re} \iint_{D \times D} \frac{u(x) \overline{v(y)}}{|Le - x - y|^\kappa} dx dy = 2 \operatorname{Re} \langle u, B_L v \rangle. \end{aligned}$$

The square terms therefore cancel, leaving no additional diagonal contribution. This proves the block representation in the sense of closed forms. Since B_L is bounded, it also gives the stated operator representation. The unitary change of variables

$$(u, v) \mapsto 2^{-1/2}(u + v, u - v)$$

yields (2.4). ■

3. The doublet splitting

We first record the elementary perturbation estimate used for the two scalar branches.

Lemma 3.1 (*Ground-state perturbation bound*). *Let A be self-adjoint with compact resolvent, simple lowest eigenvalue μ , and normalized ground state ϕ . Define the spectral gap by $g = \lambda_2(A) - \mu > 0$. Let W be bounded and self-adjoint, put $b = \|W\|$ and $w = \langle \phi, W\phi \rangle$, and suppose $b \leq g/4$. Then*

$$-\frac{2b^2}{g} \leq \lambda_1(A + W) - (\mu + w) \leq 0. \quad (3.1)$$

Proof. Let $\mathfrak{a}$ be the closed quadratic form associated with A . The upper bound follows by testing the Rayleigh quotient with ϕ . For a normalized vector $\psi = a\phi + \eta$ in the form domain of $\mathfrak{a}$, where $\eta \perp \phi$ and $t = \|\eta\|$, the spectral gap gives

$$\mathfrak{a}[\psi] \geq \mu + gt^2.$$

Since $|a|^2 = 1 - t^2$, self-adjointness gives

$$\langle \psi, W\psi \rangle - w = -t^2w + 2\operatorname{Re}\langle a\phi, W\eta \rangle + \langle \eta, W\eta \rangle.$$

Using $|w| \leq b$, $|a| \leq 1$, and $\|W\| = b$,

$$\begin{aligned} \mathfrak{a}[\psi] + \langle \psi, W\psi \rangle - (\mu + w) &\geq gt^2 - 2bt^2 - 2bt \\ &= (g - 2b)t^2 - 2bt \\ &= (g - 2b) \left(t - \frac{b}{g - 2b} \right)^2 - \frac{b^2}{g - 2b} \\ &\geq -\frac{b^2}{g - 2b} \geq -\frac{2b^2}{g}. \end{aligned}$$

Taking the infimum over normalized ψ proves the lower bound. ■

The next lemma extracts the interaction coefficient. In the reflected coordinates used here, evenness of the ground state removes the linear Taylor term. Section 4 obtains the same error order without symmetry from an algebraic cancellation in translated coordinates.

Lemma 3.2 (*Interaction-kernel expansion*). *Let*

$$I_L = \iint_{D \times D} \frac{\phi_1(x)\phi_1(y)}{|Le - x - y|^\kappa} dx dy.$$

For $L \geq 4R$,

$$|I_L - m_1^2 L^{-\kappa}| \leq C_{\kappa,R} m_1^2 L^{-\kappa-2}, \quad C_{\kappa,R} = 2^{\kappa+3} \kappa(\kappa+1) R^2. \quad (3.2)$$

Moreover,

$$\|B_L\| \leq \beta_L, \quad \beta_L = c_{d,s} 2^\kappa |D| L^{-\kappa}. \quad (3.3)$$

Proof. Set $F_L(z) = |Le - z|^{-\kappa}$. For $|z| \leq 2R$ and $L \geq 4R$, the segment from 0 to z remains at distance at least $L/2$ from Le . Put $n_z = (Le - z)/|Le - z|$. Direct differentiation gives

$$\nabla^2 F_L(z) = \kappa |Le - z|^{-\kappa-2} ((\kappa + 2)n_z \otimes n_z - I).$$

Its radial and transverse eigenvalues are respectively $\kappa(\kappa + 1)|Le - z|^{-\kappa-2}$ and $-\kappa|Le - z|^{-\kappa-2}$. Since $\kappa > 1$,

$$\|\nabla^2 F_L(z)\| = \kappa(\kappa + 1)|Le - z|^{-\kappa-2}.$$

Taylor's theorem therefore yields

$$|F_L(z) - L^{-\kappa} - \kappa L^{-\kappa-1} e \cdot z| \leq C_{\kappa,R} L^{-\kappa-2}$$

for $|z| \leq 2R$, with the displayed value of $C_{\kappa,R}$. Because ϕ_1 is even,

$$\int_D x \phi_1(x) dx = 0.$$

Consequently the integral of the linear Taylor term with $z = x + y$ vanishes. Since ϕ_1 is positive, integration of the remainder gives (3.2).

Finally, $|Le - x - y| \geq L - 2R \geq L/2$. The Schur test applied to (2.3) gives

$$\|B_L\| \leq c_{d,s} |D| (L - 2R)^{-\kappa} \leq c_{d,s} 2^\kappa |D| L^{-\kappa},$$

which is (3.3). ■

Theorem 3.3 (*Two-well algebraic splitting*). *Let $D \subset B_R(0)$ be a bounded centrally symmetric Lipschitz domain, and let Ω_L be as in Section 2. Set*

$$L_0 = \max \left\{ 4R, \left(\frac{4c_{d,s} 2^\kappa |D|}{g} \right)^{1/\kappa}, 2 \left(\frac{2c_{d,s} 4^\kappa |D|^2}{gm_1^2} \right)^{1/\kappa} \right\}. \quad (3.4)$$

For $L \geq L_0$, the first two eigenvalues of A_{Ω_L} satisfy

$$|\lambda_1(\Omega_L) - (\mu_1 - c_{d,s} m_1^2 L^{-\kappa})| \leq c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}, \quad (3.5)$$

$$|\lambda_2(\Omega_L) - (\mu_1 + c_{d,s} m_1^2 L^{-\kappa})| \leq c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}. \quad (3.6)$$

Both eigenvalues are simple, and

$$\lambda_3(\Omega_L) \geq \mu_1 + \frac{3g}{4}, \quad \lambda_3(\Omega_L) - \lambda_2(\Omega_L) \geq \frac{g}{2}. \quad (3.7)$$

In particular, as $L \rightarrow \infty$,

$$L^{d+2s} (\lambda_2(\Omega_L) - \lambda_1(\Omega_L)) \longrightarrow 2c_{d,s} m_1^2. \quad (3.8)$$

Proof. By Lemma 2.1, the spectrum is the union of the spectra of $A_D + B_L$ and $A_D - B_L$. Put

$$t_L = \langle \phi_1, B_L \phi_1 \rangle = -c_{d,s} I_L < 0.$$

For $L \geq L_0$, the second term in (3.4) gives the first inequality in

$$\beta_L \leq \frac{g}{4}, \quad 2c_{d,s} m_1^2 (L + 2R)^{-\kappa} > \frac{4\beta_L^2}{g}. \quad (3.9)$$

Indeed, $L \geq 4R$ gives $L + 2R \leq 3L/2$, while the third term in (3.4) and $3/2 < 2$ give

$$L^\kappa > \left(\frac{3}{2}\right)^\kappa \frac{2c_{d,s} 4^\kappa |D|^2}{gm_1^2}.$$

Substitution of $\beta_L = c_{d,s} 2^\kappa |D| L^{-\kappa}$ proves the strict second inequality in (3.9).

Apply Lemma 3.1 first with $W = B_L$ and then with $W = -B_L$. The two branch ground-state energies have the form

$$E_+(L) = \mu_1 + t_L + \rho_+(L), \quad E_-(L) = \mu_1 - t_L + \rho_-(L), \quad (3.10)$$

where

$$|\rho_\pm(L)| \leq \frac{2\beta_L^2}{g}.$$

The second eigenvalue of either branch is at least $\mu_2 - \beta_L \geq \mu_1 + 3g/4$, whereas both branch ground-state energies are at most $\mu_1 + \beta_L \leq \mu_1 + g/4$. Thus the first two eigenvalues of the full operator are the two numbers in (3.10).

Since $|Le - x - y| \leq L + 2R$ and $\phi_1 > 0$,

$$I_L \geq m_1^2 (L + 2R)^{-\kappa}.$$

The second condition in (3.9) and (3.10) imply $E_+(L) < E_-(L)$. Hence $\lambda_1(\Omega_L) = E_+(L)$ and $\lambda_2(\Omega_L) = E_-(L)$. The within-branch separation above makes both eigenvalues simple. It also gives $\lambda_3(\Omega_L) \geq \mu_1 + 3g/4$, while $\lambda_2(\Omega_L) \leq \mu_1 + g/4$; this proves (3.7). Equations (3.5) and (3.6) now follow from Lemma 3.2. Subtracting the two expansions and multiplying by L^κ proves (3.8), since $\kappa > 0$. $\blacksquare$

Corollary 3.4 (*Rate for the distant-ball minimizing sequence*). Let $D = B_R(0)$ and let Ω_L be the union of two radius- R balls whose centers are separated by L , so that $|\Omega_L| = 2|B_R|$. Then, as $L \rightarrow \infty$,

$$L^{d+2s} (\lambda_2(\Omega_L) - \lambda_1(B_R)) \longrightarrow c_{d,s} \left(\int_{B_R} \phi_1(x) dx \right)^2.$$

Thus the minimizing sequence in [BP16, Theorem 6.2] approaches its limiting value at the explicit leading order L^{-d-2s} .

Proof. This is (3.6) with $D = B_R(0)$ and $\mu_1 = \lambda_1(B_R)$. $\blacksquare$

4. Finitely many wells

The two-well central-inversion decomposition is special to a pair. For finitely many components we instead keep the full block operator and compare its lowest spectral cluster with its compression to the translated one-well ground states.

Throughout this section, let $D \subset B_R(0)$ be any bounded Lipschitz domain; central symmetry is not assumed. The compactness and ground-state results [DNPV12, Theorems 5.4 and 7.1] and [BP16, Theorem 2.8] apply unchanged. Write $\mu_1 < \mu_2$, $g = \mu_2 - \mu_1 > 0$, choose the positive normalized ground state $A_D \phi_1 = \mu_1 \phi_1$, and set $m_1 = \int_D \phi_1(x) dx > 0$.

Definition 4.1 (Multi-well geometry). Fix an integer $N \geq 2$ and distinct points $\mathbf{a} = (a_1, \dots, a_N)$ in $\mathbb{R}^d$. Set

$$\delta_{\mathbf{a}} = \min_{i \neq j} |a_i - a_j|, \quad \Sigma_p(\mathbf{a}) = \max_{1 \leq i \leq N} \sum_{j \neq i} |a_i - a_j|^{-p} \quad (p > 0). \quad (4.1)$$

For $L\delta_{\mathbf{a}} > 2R$, define

$$D_{j,L} = D + La_j, \quad \Omega_{\mathbf{a},L} = \bigcup_{j=1}^N D_{j,L}. \quad (4.2)$$

The sets in this union are pairwise disjoint.

On $\mathcal{H}_N = \bigoplus_{j=1}^N L^2(D)$, let $\mathcal{A}_0 = \bigoplus_{j=1}^N A_D$. For $i \neq j$ define

$$(B_{ij,L}v)(x) = -c_{d,s} \int_D \frac{v(y)}{|L(a_i - a_j) + x - y|^\kappa} dy, \quad B_{ii,L} = 0, \quad (4.3)$$

and let $\mathcal{V}_L = (B_{ij,L})_{i,j=1}^N$.

Lemma 4.2 (Exact multi-well reduction and norm bound). If $L\delta_{\mathbf{a}} \geq 4R$, then $A_{\Omega_{\mathbf{a},L}}$ is unitarily equivalent to $\mathcal{A}_0 + \mathcal{V}_L$. Moreover,

$$\|\mathcal{V}_L\| \leq \gamma_L, \quad \gamma_L = c_{d,s} 2^\kappa |D| \Sigma_\kappa(\mathbf{a}) L^{-\kappa}. \quad (4.4)$$

Proof. For $f \in L^2(\Omega_{\mathbf{a},L})$, put $u_j(x) = f(x + La_j)$ for $x \in D$. This is a unitary identification with $\mathcal{H}_N$. Translation invariance gives the diagonal operator $\mathcal{A}_0$. The cutoff argument in the proof of Lemma 2.1 identifies the form domain with $\bigoplus_{j=1}^N \tilde{H}^s(D)$, and positive separation makes every cross-kernel bounded. For $i < j$, direct expansion of the form (2.1) gives the cross term

$$-2c_{d,s} \operatorname{Re} \iint_{D \times D} \frac{u_i(x) \overline{u_j(y)}}{|L(a_i - a_j) + x - y|^\kappa} dx dy.$$

Since $B_{ji,L} = B_{ij,L}^*$, summing these terms proves the exact block representation in the sense of closed forms and hence as operators.

For $i \neq j$ and $x, y \in D$,

$$|L(a_i - a_j) + x - y| \geq L|a_i - a_j| - 2R \geq \frac{L}{2}|a_i - a_j|.$$

The Schur test therefore gives

$$\|B_{ij,L}\| \leq c_{d,s} 2^\kappa |D| L^{-\kappa} |a_i - a_j|^{-\kappa}. \quad (4.5)$$

For $u = (u_1, \dots, u_N)$, set $r_j = \|u_j\|_{L^2(D)}$ and let $\mathbf{N}_L$ be the symmetric matrix with entries $(\mathbf{N}_L)_{ij} = \|B_{ij,L}\|$. Then

$$\|\mathcal{V}_L u\|_{\mathcal{H}_N} \leq \|\mathbf{N}_L r\|_{\ell^2} \leq \left(\max_i \sum_j (\mathbf{N}_L)_{ij} \right) \|r\|_{\ell^2}.$$

Combining this inequality with (4.5) proves (4.4). $\blacksquare$

We next record the finite-multiplicity perturbation estimate that replaces Lemma 3.1.

Lemma 4.3 (*Isolated spectral-cluster bound*). *Let A be self-adjoint with compact resolvent. Suppose its lowest eigenvalue μ has multiplicity N , let P be the corresponding spectral projection, and suppose*

$$A|_{P^\perp} \geq \mu + g$$

in the quadratic-form sense for some $g > 0$. Let W be bounded and self-adjoint, set $b = \|W\|$, and assume $b \leq g/4$. If $\eta_1 \leq \dots \leq \eta_N$ are the eigenvalues of PWP on $\text{ran } P$, then, for $1 \leq k \leq N$,

$$-\frac{2b^2}{g} \leq \lambda_k(A + W) - (\mu + \eta_k) \leq 0. \quad (4.6)$$

In addition,

$$\lambda_{N+1}(A + W) \geq \mu + g - b \geq \mu + \frac{3g}{4}. \quad (4.7)$$

Proof. The min-max principle applied to the span of the first k eigenvectors of PWP gives the upper bound in (4.6). For the lower bound, write $\psi = p + q$ with $p = P\psi$ and $q = (I - P)\psi$. Then

$$\begin{aligned} \langle \psi, (A + W)\psi \rangle &\geq \mu \|\psi\|^2 + \langle p, PWPp \rangle + (g - b)\|q\|^2 - 2b\|p\|\|q\| \\ &\geq \mu \|\psi\|^2 + \langle p, PWPp \rangle - \frac{2b^2}{g}\|p\|^2 + \left(\frac{g}{2} - b\right)\|q\|^2. \end{aligned}$$

The second line uses $2b\|p\|\|q\| \leq 2b^2g^{-1}\|p\|^2 + (g/2)\|q\|^2$. Thus, in the form sense, $A + W$ dominates the direct sum of

$$\mu I + PWP - \frac{2b^2}{g}I \quad \text{on } \text{ran } P$$

and $(\mu + g/2 - b)I$ on $P^\perp$. Since $\eta_N \leq b \leq g/4$, we have

$$\eta_N - \frac{2b^2}{g} \leq b - \frac{2b^2}{g} < \frac{g}{2} - b.$$

The strict inequality follows because the difference between the last two quantities is $(g - 2b)^2/(2g) > 0$. Hence the first N eigenvalues of this comparison operator are the N eigenvalues of its P -block. Another application of the min-max principle gives the lower bound in (4.6). Finally, $A + W \geq A - bI$ and the min-max principle give (4.7). $\blacksquare$

Let P_1 be the orthogonal projection in $\mathcal{H}_N$ onto the span of the N vectors whose j th component is ϕ_1 and whose other components vanish. In this basis the compression $P_1 \mathcal{V}_L P_1$ is represented by the real symmetric matrix $T_L = (t_{ij}(L))$, where

$$t_{ii}(L) = 0, \quad t_{ij}(L) = -c_{d,s} \iint_{D \times D} \frac{\phi_1(x)\phi_1(y)}{|L(a_i - a_j) + x - y|^\kappa} dx dy \quad (i \neq j). \quad (4.8)$$

Definition 4.4 (Effective interaction matrix). Define the real symmetric $N \times N$ matrix $M_{\mathbf{a}}$ by

$$(M_{\mathbf{a}})_{ii} = 0, \quad (M_{\mathbf{a}})_{ij} = -c_{d,s} m_1^2 |a_i - a_j|^{-\kappa} \quad (i \neq j). \quad (4.9)$$

Write its ordered eigenvalues as $\theta_1 \leq \dots \leq \theta_N$.

Lemma 4.5 (Compression error). If $L\delta_{\mathbf{a}} \geq 4R$, then

$$\|T_L - L^{-\kappa} M_{\mathbf{a}}\|_{\ell^2 \rightarrow \ell^2} \leq \varepsilon_L, \quad \varepsilon_L = c_{d,s} C_{\kappa,R} m_1^2 \Sigma_{\kappa+2}(\mathbf{a}) L^{-\kappa-2}. \quad (4.10)$$

Proof. Fix $i \neq j$, put $a = a_i - a_j$, and set $F(z) = |La + z|^{-\kappa}$. For $|z| \leq 2R$, the segment from 0 to z stays at distance at least $L|a|/2$ from $-La$. As in the proof of Lemma 3.2, Taylor's theorem gives

$$|F(z) - F(0) - \nabla F(0) \cdot z| \leq C_{\kappa,R} L^{-\kappa-2} |a|^{-\kappa-2}. \quad (4.11)$$

The linear term vanishes after integration because

$$\iint_{D \times D} \phi_1(x)\phi_1(y)(x - y) dx dy = 0.$$

This cancellation is algebraic and does not require central symmetry of D . Using $F(0) = L^{-\kappa} |a|^{-\kappa}$ in (4.8) and integrating (4.11) gives

$$|t_{ij}(L) - L^{-\kappa} (M_{\mathbf{a}})_{ij}| \leq c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} |a_i - a_j|^{-\kappa-2}.$$

The operator norm of a real symmetric matrix is at most its largest absolute row sum. Taking that row sum proves (4.10). $\blacksquare$

Theorem 4.6 (Finite multi-well effective Hamiltonian). Let D , μ_1 , g , ϕ_1 , and m_1 be as fixed at the beginning of this section, and let the multi-well geometry be as in Definition 4.1. If

$$L\delta_{\mathbf{a}} \geq 4R, \quad \gamma_L \leq \frac{g}{4}, \quad (4.12)$$

then, for $1 \leq k \leq N$,

$$|\lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + L^{-\kappa} \theta_k)| \leq \varepsilon_L + \frac{2\gamma_L^2}{g}. \quad (4.13)$$

The upper edge of the cluster and the next eigenvalue satisfy

$$\lambda_N(\Omega_{\mathbf{a},L}) \leq \mu_1 + \gamma_L \leq \mu_1 + \frac{g}{4}, \quad \lambda_{N+1}(\Omega_{\mathbf{a},L}) \geq \mu_1 + g - \gamma_L \geq \mu_1 + \frac{3g}{4}. \quad (4.14)$$

In particular, the cluster gap has the quantitative lower bound

$$\lambda_{N+1}(\Omega_{\mathbf{a},L}) - \lambda_N(\Omega_{\mathbf{a},L}) \geq g - 2\gamma_L \geq \frac{g}{2}. \quad (4.15)$$

Consequently, as $L \rightarrow \infty$, for every $1 \leq k \leq N$,

$$L^{d+2s}(\lambda_k(\Omega_{\mathbf{a},L}) - \mu_1) \rightarrow \theta_k. \quad (4.16)$$

Proof. Lemma 4.2 identifies the full operator with $\mathcal{A}_0 + \mathcal{V}_L$. The lowest eigenvalue of $\mathcal{A}_0$ is μ_1 , with multiplicity N , its spectral projection is P_1 , and the rest of its spectrum lies in $[\mu_1 + g, \infty)$. Apply Lemma 4.3 with $A = \mathcal{A}_0$, $W = \mathcal{V}_L$, and $b = \|\mathcal{V}_L\| \leq \gamma_L$. If $\tau_1(L) \leq \dots \leq \tau_N(L)$ are the eigenvalues of T_L , this gives

$$-\frac{2\gamma_L^2}{g} \leq \lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + \tau_k(L)) \leq 0. \quad (4.17)$$

The finite-dimensional min-max principle and Lemma 4.5 give

$$|\tau_k(L) - L^{-\kappa}\theta_k| \leq \varepsilon_L.$$

Combining these two estimates proves (4.13), and the upper bound in (4.17), together with $\tau_N(L) \leq \|T_L\| \leq \|\mathcal{V}_L\| \leq \gamma_L$, gives the upper-edge estimate in (4.14). The lower bound for λ_{N+1} follows from (4.7). Subtracting the two estimates proves (4.15). Finally, $L^\kappa \varepsilon_L = O(L^{-2})$ and $L^\kappa \gamma_L^2 = O(L^{-\kappa})$; since $\kappa = d + 2s > 0$, multiplying (4.13) by L^κ proves (4.16). ■

Corollary 4.7 (*Symmetry-free two-well asymptotic*). Let $D \subset B_R(0)$ be any bounded Lipschitz domain, let $a_1 \neq a_2$, and put $r = |a_1 - a_2|$. For $\Omega_L = (D + La_1) \cup (D + La_2)$, as $L \rightarrow \infty$,

$$\begin{aligned} \lambda_1(\Omega_L) &= \mu_1 - c_{d,s} m_1^2 r^{-\kappa} L^{-\kappa} + O(L^{-\kappa-2} + L^{-2\kappa}), \\ \lambda_2(\Omega_L) &= \mu_1 + c_{d,s} m_1^2 r^{-\kappa} L^{-\kappa} + O(L^{-\kappa-2} + L^{-2\kappa}). \end{aligned}$$

In particular,

$$L^{d+2s}(\lambda_2(\Omega_L) - \lambda_1(\Omega_L)) \rightarrow 2c_{d,s} m_1^2 r^{-d-2s}.$$

Proof. For $N = 2$, the effective matrix is

$$M_{\mathbf{a}} = \begin{pmatrix} 0 & -c_{d,s} m_1^2 r^{-\kappa} \\ -c_{d,s} m_1^2 r^{-\kappa} & 0 \end{pmatrix},$$

whose ordered eigenvalues are $-c_{d,s} m_1^2 r^{-\kappa}$ and $c_{d,s} m_1^2 r^{-\kappa}$. Theorem 4.6 gives both expansions because $\varepsilon_L = O(L^{-\kappa-2})$ and $\gamma_L^2 = O(L^{-2\kappa})$. Subtracting them and using $\kappa = d + 2s$ gives the limit. ■

Corollary 4.8 (*Recovery of the doublet*). Assume in addition that $D = -D$. For $N = 2$, $a_1 = -e/2$, and $a_2 = e/2$, the right-hand side of (4.13), including its explicit constants, agrees with the bounds (3.5)–(3.6).

Proof. In this case

$$M_{\mathbf{a}} = \begin{pmatrix} 0 & -c_{d,s}m_1^2 \\ -c_{d,s}m_1^2 & 0 \end{pmatrix},$$

whose ordered eigenvalues are the two stated numbers. Moreover, $\Sigma_\kappa(\mathbf{a}) = \Sigma_{\kappa+2}(\mathbf{a}) = 1$, so $\gamma_L = \beta_L$. In the exact compression, central symmetry of D and evenness of ϕ_1 show, after the change of variables $y \mapsto -y$, that the translated two-well kernel integral equals I_L . Thus the compression errors, and hence the two displayed remainder bounds, coincide. ■

Corollary 4.9 (*Collective ground state*). *For every multi-well configuration, $\theta_1 < 0$ is simple and admits an eigenvector with strictly positive entries, while $\theta_N > 0$. For every L admitted by Definition 4.1, $\lambda_1(\Omega_{\mathbf{a},L})$ is simple. Moreover, for all sufficiently large L , $\lambda_1(\Omega_{\mathbf{a},L}) < \mu_1 < \lambda_N(\Omega_{\mathbf{a},L})$.*

Proof. The off-diagonal entries of $M_{\mathbf{a}}$ are strictly negative, and its Rayleigh quotient at $(1, \dots, 1)$ is negative, so $\theta_1 < 0$. Replacing a Rayleigh minimizer z by its componentwise absolute value cannot increase the quotient; equality forces its nonzero components to have the same sign, and the eigenvalue equation rules out a zero component. Thus a ground-state eigenvector has one strict sign. If the ground eigenspace had dimension at least two, it would contain a nonzero vector with a zero component, a contradiction. Hence θ_1 is simple. Finally, $\text{tr } M_{\mathbf{a}} = 0$ gives $\theta_N > 0$. Put $r_L = \varepsilon_L + 2\gamma_L^2/g$. Theorem 4.6 gives $L^\kappa r_L \rightarrow 0$. Simplicity of $\lambda_1(\Omega_{\mathbf{a},L})$ for every admissible L follows from [BP16, Theorem 2.8], applied to the bounded open set $\Omega_{\mathbf{a},L}$. Choose L so large that

$$L^\kappa r_L < \min\{-\theta_1, \theta_N\}.$$

Then (4.13) gives $\lambda_1 < \mu_1 < \lambda_N$, which proves the remaining assertions. ■

Corollary 4.10 (*Regular-simplex cluster*). *Suppose the centers satisfy $|a_i - a_j| = r$ for all $i \neq j$, as for the vertices of a regular simplex. Then $N \leq d + 1$. Put*

$$w = c_{d,s}m_1^2r^{-\kappa}.$$

Then $\theta_1 = -(N-1)w$ and $\theta_2 = \dots = \theta_N = w$. Thus one collective level is lowered by $(N-1)wL^{-\kappa}$, while the remaining $N-1$ levels are raised by $wL^{-\kappa}$, up to the remainder in (4.13).

Proof. After translating a_N to the origin, the $N-1$ vectors $a_i - a_N$ have Gram matrix with diagonal entries r^2 and off-diagonal entries $r^2/2$. This matrix is positive definite, so $N-1 \leq d$. Here $M_{\mathbf{a}} = -w(J - I)$, where J is the all-ones matrix. The vector $(1, \dots, 1)$ has eigenvalue $-(N-1)w$, and its orthogonal complement has eigenvalue w . ■

5. A reproducible interval check

We finish with a conforming fixed-mesh Galerkin check of the two- and three-well limits. It tests the formulas and is not an input to their proofs. No $h \rightarrow 0$ discretization-error estimate is claimed; the calculation tests the proved $L \rightarrow \infty$ asymptotic at fixed mesh. Take $d = 1$,

$0 < s < 1/2$, and consider finite unions of unit intervals. Partition each component into cells of length h . The cell indicators have finite form energy by Proposition 5.1, so they span a conforming finite-dimensional subspace of the form domain.

Proposition 5.1 (*Exact cell-integral matrix*). Assume $0 < s < 1/2$. Let $I_1, \dots, I_M$ be pairwise disjoint cells of common length h , with centers x_α , and set $G = \bigcup_{\alpha=1}^M I_\alpha$, $r_{\alpha\beta} = |x_\alpha - x_\beta|$, and $q = 1 - 2s$. In the $L^2(G)$ -orthonormal basis $e_\alpha = h^{-1/2} \mathbf{1}_{I_\alpha}$, the Galerkin matrix H_h of the zero-exterior form Π_G has entries

$$(H_h)_{\alpha\alpha} = \frac{c_{1,s}}{h} \frac{2h^q}{2sq}, \quad (5.1)$$

$$(H_h)_{\alpha\beta} = -\frac{c_{1,s}}{h} \frac{2r_{\alpha\beta}^q - (r_{\alpha\beta} + h)^q - (r_{\alpha\beta} - h)^q}{2sq}, \quad \alpha \neq \beta. \quad (5.2)$$

Proof. For two disjoint cells with center distance $r \geq h$, two direct integrations give

$$\iint_{I_\alpha \times I_\beta} |x - y|^{-1-2s} dx dy = \frac{2r^q - (r + h)^q - (r - h)^q}{2sq}.$$

The off-diagonal form entry is minus $c_{1,s}$ times this integral. For one cell,

$$\int_{I_\alpha} \int_{\mathbb{R} \setminus I_\alpha} |x - y|^{-1-2s} dy dx = \frac{2h^q}{2sq}.$$

Dividing both expressions by h for the normalized indicators proves (5.1)–(5.2). ■

Corollary 5.2 (*Fixed-mesh cluster asymptotic*). Let $D = (-1/2, 1/2)$, and fix a uniform partition of D . Let $\mu_{1,h}$ and $\phi_{1,h}$ be the lowest eigenvalue and a positive, L^2 -normalized piecewise-constant eigenfunction of the one-well Galerkin matrix. Put $m_{1,h} = \int_D \phi_{1,h}$. Translate this same partition to each component $D + La_j$, and write the ordered eigenvalues of the resulting Galerkin matrix on $\bigcup_{j=1}^N (D + La_j)$ as $\lambda_{k,h}(L)$. Then, as $L \rightarrow \infty$, for fixed distinct $a_1, \dots, a_N$,

$$\lambda_{k,h}(L) = \mu_{1,h} + L^{-1-2s} \theta_{k,h} + O_{h,\mathbf{a},s}(L^{-3-2s} + L^{-2-4s}), \quad 1 \leq k \leq N, \quad (5.3)$$

where $\theta_{1,h} \leq \dots \leq \theta_{N,h}$ are the eigenvalues of the matrix with zero diagonal and off-diagonal entries $-c_{1,s} m_{1,h}^2 |a_i - a_j|^{-1-2s}$.

Proof. All off-diagonal entries of the one-well matrix in (5.2) are negative. The Rayleigh-quotient argument in the proof of Corollary 4.9 therefore gives a simple lowest eigenvalue and a strictly positive eigenvector. On the direct sum of N fixed Galerkin spaces, the block expansion used in Lemma 4.2 is exact, its off-diagonal norm is $O_{h,\mathbf{a},s}(L^{-1-2s})$, and the complement of the N copied ground states is separated by the positive discrete one-well gap. The Taylor argument in Lemma 4.5, now in Euclidean operator norm, gives an $O_{h,\mathbf{a},s}(L^{-3-2s})$ compression error. Applying Lemma 4.3 to this finite matrix gives the additional $O_{h,\mathbf{a},s}(L^{-2-4s})$ term and proves (5.3). ■

For the computation we take $s = 1/4$, $D = (-1/2, 1/2)$, and the standard normalization

$$c_{1,s} = \frac{4^s s \Gamma(1/2 + s)}{\sqrt{\pi} \Gamma(1 - s)}.$$

Table 1 records the one-well quantities. The stabilization of $m_{1,h}$ is the relevant check for the interaction coefficient.

cells	$\mu_{1,h}$	$\mu_{2,h} - \mu_{1,h}$	$m_{1,h}$
60	1.3794549573	0.9105539931	0.9628386433
120	1.3753692130	0.8998883318	0.9633505837
180	1.3741292905	0.8969647744	0.9636021778

Table 1. One-well Galerkin data for $s = 1/4$.

On the 180-cell mesh, the two-well component centers are $(-L/2, L/2)$, while the three-well centers are $(-L, 0, L)$. The corresponding fixed-mesh effective spectra are

$$\begin{aligned} (-L/2, L/2) : & \quad (-0.18521477, 0.18521477), \\ (-L, 0, L) : & \quad (-0.29671332, 0.06548331, 0.23123001). \end{aligned}$$

The direct Galerkin eigenvalues are shown in Table 2; the last column is the largest absolute difference from the corresponding effective-matrix eigenvalue. These errors are computed from the full-precision values before the shifts and effective spectra are rounded for display.

L	$L^{3/2}(\lambda_{1,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{2,h} - \mu_{1,h})$		max. error
3	-0.190740	0.189832		5.52×10^{-3}
6	-0.186544	0.186335		1.33×10^{-3}
12	-0.185551	0.185486		3.36×10^{-4}
24	-0.185302	0.185279		8.68×10^{-5}

L	$L^{3/2}(\lambda_{1,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{2,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{3,h} - \mu_{1,h})$	max. error
3	-0.304984	0.065562	0.237544	8.27×10^{-3}
6	-0.298738	0.065554	0.232747	2.03×10^{-3}
12	-0.297236	0.065504	0.231595	5.22×10^{-4}
24	-0.296852	0.065488	0.231316	1.38×10^{-4}

Table 2. Scaled two-well (top) and three-well (bottom) cluster shifts at selected separations.

At the selected separations, the scaled errors decrease monotonically; Corollary 5.2 proves that they tend to zero at fixed mesh. At $L = 3$, the discrete bounds are $\gamma_L = 0.10858$ for two wells and $\gamma_L = 0.21716$ for three wells, both below $g_h/4 = 0.22424$. Since γ_L decreases with L and $L\delta_{\mathbf{a}} \geq 4R$ already at $L = 3$, the finite-dimensional cluster estimate used in the proof applies at every computed separation. The calculation uses the exact entries (5.1)–(5.2); no singular quadrature or fitted coefficient enters the tables.

AI-use disclosure

The proof architecture and manuscript were developed with assistance from OpenAI’s GPT-5.6 Sol model (`gpt-5.6-sol`). The model served as an interactive research assistant for argument development and stress testing, citation and calculation audits, preparation of reproducibility and release-audit code, and revision. The author directed the research, selected and checked the final arguments, and takes responsibility for all mathematical claims, citations, computations, and wording.

Code availability

The script `tools/check_asymptotics.py` reproduces Tables 1 and 2. Before solving the eigenproblems, it checks the assembled constant-function energy identity and the exact reuse of each translated one-well block, then prints progress after each separation. The pinned numerical dependencies are listed in `requirements-numeric.txt`; the reference output was replayed with Python 3.12.3, NumPy 1.26.4, and SciPy 1.11.4. The submission source package includes frozen copies of the scripts, pinned dependencies, and reference output; this is the version corresponding to the manuscript. The version tag for this manuscript is `paper-2026-09-04`. The development repository is available at github.com/skypher/laplace-tunneling.

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